

# Zeeshan Manzoor Bhat

Research Associate | Structural & Earthquake Engineering

Department of Earthquake Engineering  
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## RESEARCH INTERESTS

Large-scale experimental testing; Nonlinear modelling of structures; Seismic hazard, risk and vulnerability assessment; Performance-based earthquake engineering; Resilience-based seismic design.

## EDUCATION

### Ph.D. in Structural and Earthquake Engineering

July 2018 – March 2025

Indian Institute of Technology Roorkee, India

**Thesis:** Seismic Performance of Masonry Infilled RC Frame Buildings Designed for Modern Codes

**Advisors:** [Prof. Yogendra Singh](#) & [Prof. Pankaj Agarwal](#)

### M.Tech. in Structural Engineering

July 2016 – July 2018

National Institute of Technology Srinagar, India

**Thesis:** An Experimental Investigation into the Behavior of Steel–Timber Composite Beams

**Advisor:** [Prof. Javed Ahmad Bhat](#)

### B.Tech. in Civil Engineering

July 2012 – June 2016

National Institute of Technology Srinagar, India

## RESEARCH EXPERIENCE

### Research Associate

June 2025 – Present

Indian Institute of Technology Roorkee, India

**Project:** Enhancing Sustainability of RC Frame Buildings through Seismic Safety and Design Life Elongation Using GFRP Reinforcement

## FUNDED PROJECTS

### Next-Generation Climate-Resilient and Energy-Efficient Housing

Queen's University, Kingston, Canada

**Funding:** \$140,000 through [Canada Impact + Research Training Awards](#)

**Supervisor:** [Prof. Amir Fam](#)

## PUBLICATIONS

### Peer-Reviewed Journal Articles

- [1] Bhat, Z. M., & Singh, Y. (2026). Out-of-plane seismic fragility analysis of masonry infills in low- and mid-rise RC frame archetype buildings accounting for in-plane damage. *Journal of Building Engineering*. [Link](#)
- [2] Bhat, Z. M., & Singh, Y. (2025). Seismic risk assessment of fly ash brick and AAC block masonry infilled RC frame buildings designed for modern codes. *Journal of Earthquake Engineering*. [Link](#)
- [3] Bhat, Z. M., & Singh, Y. (2025). Out-of-plane behavior of masonry infills of different types and slenderness ratios under reversed cyclic loading. *Journal of Structural Engineering*. [Link](#)
- [4] Bhat, Z. M., Singh, Y., & Agarwal, P. (2024). Cyclic testing and diagonal strut modelling of different types of masonry infills in reinforced concrete frames designed for modern codes. *Engineering Structures*. [Link](#)
- [5] Bhat, Z. M., Singh, Y., & Agarwal, P. (2023). Characterization of mechanical behavior of different types of masonry with full-field strain using digital image correlation. *Construction and Building Materials*. [Link](#)
- [6] Waseem, S. A., Bhat, Z. M., & Bhat, J. A. (2022). Experimental investigation into the behavior of steel–timber composite beams. *Practice Periodical on Structural Design and Construction*. [Link](#)

## Under Review

- [7] **Bhat, Z. M., & Singh, Y.** Full-scale cyclic testing and DIC-based damage assessment of masonry infilled RC frames with openings. *ACI Structural Journal*.

## INTERNATIONAL CONFERENCES

- [1] **Bhat, Z. M., & Singh, Y.** (2025). Reversed out-of-plane cyclic testing of masonry infills using double-leaf airbags. *COMPDYN 2025: 10th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering*, Rhodes Island, Greece, 15–18 June 2025. [Link](#)
- [2] **Bhat, Z. M., & Singh, Y.** (2025). Cyclic in-plane behavior of masonry infilled RC frames with detailed interpretation of damage using digital image correlation. *15th Canadian Masonry Symposium*, Ottawa, Canada, 2–5 June 2025.

## TECHNICAL REPORTS

- [1] **Bhat, Z. M., Singh, Y., & Agarwal, P.** (2024). *Seismic Safety Measures for RC Frame Buildings with Different Types of Infill Panels*. NBCC (India) Limited, India.
- [2] Team Member. (2022). *Compendium of Traditional Earthquake-Resilient Construction Practices*. NDMA, Government of India. [Link](#)

## OPEN DATASETS

- [1] **Bhat, Z. M., & Singh, Y.** (2024). Fly ash brick and AAC block masonry infilled RC frame buildings: OpenSees models and responses. *DesignSafe-CI*. [Link](#)

## AWARDS AND ACHIEVEMENTS

- Canada Impact + Research Training Award – Postdoctoral, **Natural Sciences and Engineering Research Council of Canada (NSERC)**, 2026
- Research Fellowship, **Sapienza University of Rome, Italy**, 2025
- MHRD Fellowship for Doctoral Studies, **Ministry of Human Resource Development, Government of India**
- MHRD Fellowship for M.Tech Studies, **Ministry of Human Resource Development, Government of India**
- Ranked among the top 2% in the **Graduate Aptitude Test in Engineering (GATE)** 2016, among 119,873 candidates

## TECHNICAL SKILLS

- **Programming & Tools:** Python, MATLAB, TCL, LaTeX
- **Software:** OpenSees, STKO, ETABS, SAP2000, OpenQuake
- **Experimental Techniques:** Digital Image Correlation (DIC), Laboratory Instrumentation and Data Acquisition, Servo-hydraulic Actuator Control, Load Cell Calibration

## INTERNATIONAL WORKSHOPS

- 6th International Course on Seismic Analysis of Structures** 19–22 July 2021  
University of Palermo, Italy
- Nonlinear Modelling of RC Structures Using OpenSees** 22–26 June 2020  
Università degli Studi di Napoli Federico II, Italy

## TEACHING ASSISTANTSHIPS

- Advanced Earthquake Resistant Design of Structures (EQN-512)** IIT Roorkee  
Instructor: [Prof. Yogendra Singh](#)
- Earthquake Resistant Design of Structures (EQN-563)** IIT Roorkee  
Instructor: [Prof. Yogendra Singh](#)

## REFERENCES

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**Prof. Yogendra Singh**

Professor, Department of Earthquake Engineering, Indian Institute of Technology Roorkee, India

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**Prof. Pankaj Agarwal**

Professor, Department of Earthquake Engineering, Indian Institute of Technology Roorkee, India

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**Prof. Pradeep Bhargava**

Professor, Department of Civil Engineering, Indian Institute of Technology Roorkee, India

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